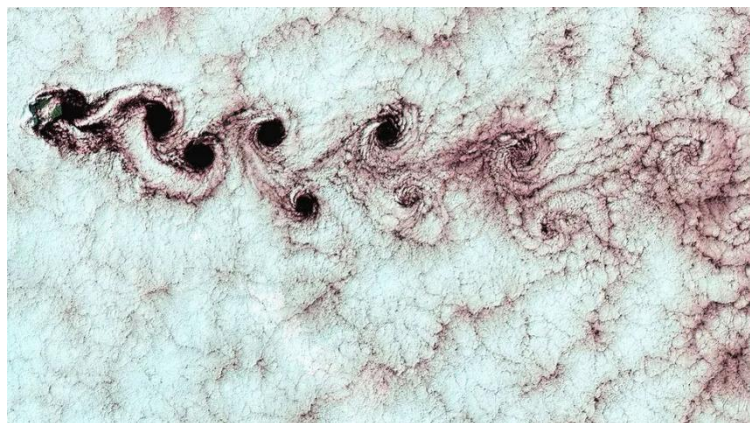


考研精读 | Economist



The Navier-Stokes problem

纳维-斯托克斯之问

OpenAI's apparent maths breakthrough raises profound questions

OpenAI 数学突破斐然，引发深刻疑问

难度评级：★★★★☆

【 Para.1 】

To mark the turn of the century the Clay Mathematics Institute, in Denver, took stock of what mathematicians knew and what they did not. The result was a set of seven Millennium Prize problems representing the most important unsolved matters in maths. Solutions would attract a **purse** of \$1m.

为纪念世纪之交，位于丹佛的克雷数学研究所对数学家们已经掌握的知识以及尚未解决的问题进行了一次盘点。最终，该研究所提出了七条“千禧年大奖难题”，代表数学领域最重要的未解之问。每破解一道难题，即可获得一百万美元的奖金。

【 Para.2 】

September 12TH-18TH | Science & technology

文章导读：

克雷研究所设立七大千禧年数学难题，每题悬赏百万美元。OpenAI 宣称依靠上万 AI 智能体，仅用 88 小时破解纳维-斯托克斯方程难题，构造出会产生奇点的流体涡旋。但该成果引发学术争议，涉及研究思路与训练数据纠纷，同时引出成果归属等深层问题，OpenAI 表示不会申领奖金。

profound adj. /prə'faʊnd/ very great or intense; having or showing great knowledge or insight 深远的；深刻的；意义重大的【考研大纲词汇】

purse n. /pɜːs/ the total amount of money that an organization or government has available for spending (机构或政府的) 备用金，可用的资金，财力【考研大纲词汇】

equation n. /i'kwɛɪʃn/ a statement showing that two amounts or values are equal 方程；方程式；等式【考研大纲词汇】

n. a problem or situation in which several things must be considered and dealt with (多种因素的) 平衡，综合体

bottle up phrasal verb to not allow other people to see that you are unhappy, angry, etc., especially when this happens over a long period of time 长时间掩饰，遏制，隐瞒（不快等）
Try not to **bottle up** your emotions. 尽量不要压抑自己的情感。

bottle it phrasal verb (BrE informal) to not do sth, or not finish sth, because you are frightened 不敢做；（中途）放弃，退缩

He should have taken the penalty but he **bottled it**. 他本来应该罚点球的，可是退缩了。

squeeze v. /skwiːz/ (~ sb/sth into, through, etc. sth) (~ through, in, past, etc.) to force sb/sth/yourself into or through a small space (使) 挤入；挤过；塞入【考研大纲词汇】

We managed to **squeeze** six people into the car. 我们在那辆车上挤进了六个人。

plasma n. /'plæzmə/ (also plasm /'plæzəm/) a gas that contains approximately equal numbers of positive and negative electric charges and is present in the sun and most stars 等离子体；等离子气体（一种几乎没有电荷的气体，发现于太阳和其他星球上）

circulate v. /'sɜːkjələrt/ when a liquid, gas, or air circulates or is circulated, it moves continuously around a place or system (液体或气体) 环流，循环【考研大纲词汇】

The condition prevents the blood from **circulating** freely. 这种病阻碍了血液的循环畅通。

v. if a story, an idea, information, etc. circulates or if you circulate it, it spreads or it is passed from one person to another 传播；流传；散布

v. (~ sth to sb) to send goods or information to all the people in a group 传送；传递；传阅

v. to move around a group, especially at a party, talking to different people (尤指在聚会上) 往来应酬，周旋

Until September 8th only one of these problems had been solved. On that day, however, OpenAI announced it had conquered another. This is called the Navier-Stokes existence and smoothness problem, and it relates to a set of **equations** that capture the flow of fluids. Mathematicians have been trying to **bottle** this one **up** for at least a century. OpenAI says a team of its AI agents solved it in just 88 hours.

截至 9 月 8 日，这七大难题中仅有一道得以解决。然而，就在这一天，OpenAI 宣布，他们又攻克了另一个。这就是所谓的“纳维 - 斯托克斯方程的存在性与光滑性问题”，它涉及一组用于描述流体运动的方程。至少一个世纪以来，数学家们一直试图攻克这一难题。OpenAI 表示，其一组 AI 智能体仅用 88 小时就解决了这个问题。

【 Para.3 】

The Navier-Stokes equations are useful for modelling how air flows over aircraft wings, oil **squeezes** through pipes and **plasma circulates** inside stars. But, though mathematicians have a good **grasp** of how to **harness** them practically, a thorough understanding of the **properties** of the equations themselves **eludes** them.

纳维 - 斯托克斯方程可用于模拟空气如何流过飞机机翼、石油如何在管道中受到挤压而流动，以及等离子体如何在恒星内部循环。然而，尽管数学家们已经很好地掌握了如何在实际应用中利用这些方程，却始终未能彻底理解这些方程本身所具有的性质。

grasp n. /grɑːsp/ a person's understanding of a subject or of difficult facts 理解 (力)；领会【考研英语大纲词汇】

n. the ability to get or achieve sth 能力所及

Success was within her **grasp**. 她有把握获得成功。

v. to take a firm hold of sb/sth 抓紧；抓牢

He **grasped** my hand and shook it warmly. 他热情地抓住我的手握了起来。

v. to understand sth completely 理解；领会；领悟；明白

It took him some time to **grasp** that he was now a public figure. 他过了些时候才意识到自己已是公众人物了。

harness v. /ˈhɑːnɪs/ to control and use the force or strength of sth to produce power or to achieve sth 控制，利用 (以产生能量等)

We must **harness** the skill and creativity of our workforce. 我们必须尽量发挥全体职工的技能 and 创造力。

property n. /ˈprɒpəti/ (usually pl.) (formal) a quality or characteristic that sth has 性质；特性【考研英语大纲词汇】

elude v. /iˈluːd/ if sth eludes you, you are not able to achieve it, or not able to remember or understand it 使达不到；使不记得；使不理解【考研英语大纲词汇】

They're a popular band but chart success has **eluded** them so far. 他们是一支很受欢迎的乐队，但到目前为止还未能每周流行唱片榜上取得佳绩。

v. to manage to avoid or escape from sb/sth, especially in a clever way (尤指机敏地) 避开，逃避，躲避

The two men managed to **elude** the police for six weeks. 这两个男人想方设法逃避警方追捕达六个星期。

malfunction v. /ˌmælˈfʌŋkʃn/ to fail to work correctly 运转失常；失灵；出现故障【考研英语大纲词汇】

The radiation can damage microprocessors and computer memories, causing them to **malfunction**. 该辐射会破坏微处理器和计算机存储器，使之出故障。

density n. /ˈdensəti/ the number of people or things in a place when compared with the size of the place (人口等的) 密度【考研英语大纲词汇】

singularity n. /ˌsɪŋɡjuːˈlærəti/ (mathematics, physics) a point at which a mathematical function or physical quantity takes an infinite value, such as at the centre of a black hole (数学、物理) 奇点；奇异点

spawn v. /spɔːn/ to cause sth to develop or be produced 引发；引起；导致；造成【考研英语大纲词汇】

The band's album **spawned** a string of hit singles. 这支乐队的专辑繁衍出一连串走红的单曲唱片。

power v. /ˈpaʊə(r)/ to supply a machine or vehicle with the energy that makes it work 驱动，推动 (机器或车辆)【考研英语大纲词汇】

The aircraft is **powered** by a jet engine. 这架飞机由喷气发动机驱动。

n. (~ over sb/sth) (~ to do sth) the ability to control people or things 控制力；影响力；操纵力

n. a country with a lot of influence in world affairs, or with great military strength 有影响力的大国；军事强国

stepping-stone n. /ˈstepɪŋ stəʊn/ (also stepping stone) an event or experience that helps you achieve something else 进身之阶；敲门砖，垫脚石，跳板

【 Para.4 】

This is where the Millennium problem comes in. It asks “whether the Navier-Stokes equations can **malfunction** in a finite amount of time when describing a fluid of fixed **density** moving in three dimensions”. Such a malfunction would come in the form of a **singularity**: a point where a quantity, such as the fluid’s speed, becomes infinite and the equations break down. So far, mathematicians have been unable to prove such singularities never exist, but have also failed to find one.

这正是“千禧年大奖难题”所要解决的问题。它要问的是：“当纳维 - 斯托克斯方程用于描述一种密度固定、在三维空间中运动的流体时，该方程是否可能在有限时间内发生失效？”如果可能的话，这种失效将以“奇点”的形式出现。奇点指在某一点上，例如流体的速度等物理量变得无穷大，从而导致方程失去作用。迄今为止，数学家既无法证明这样的奇点绝不可能存在，也未能找到一个真正的奇点。

【 Para.5 】

Now, OpenAI claims its AI agents have found one. On August 28th the firm’s scientists began training a new model. They **spawned** teams of AI agents, **powered** by this model, that can read the internet, run code and communicate with one another. On September 1st they set them to work on the open Millennium problems.

swarm n. /swɔ:m/ a large group of people, especially when they are all moving quickly in the same direction 一大群，一大批（向同方向移动的人）【考研英语词汇】

v. to move around in a large group 成群地来回移动

Tourists were **swarming** all over the island. 岛上到处是旅游者熙来攘往。

swarm with sb/sth phrasal verb to be full of people or things 到处是人（或物）；挤满

The capital city is **swarming with police**. 首都到处是警察。

autonomous adj. /ɔ:ˈtɒnəməs/ able to do things and make decisions without help from anyone else 自主的；有自主权的【考研英语词汇】

adj. able to govern itself or control its own affairs 自治的；有自治权的

They proudly declared themselves part of a new **autonomous province**. 他们自豪地宣布自己是新自治省的一部分。

swirl v. /swɜ:l/ to move around quickly in a circle; to make sth do this (使) 打旋，旋动，起旋涡【考研英语词汇】

The water **swirled** down the drain. 水打着旋流进了下水道。

vortex n. /ˈvɔ:teks/ (pl. **vortexes** or **vortices** /-tɜ:z/) (technical 术语) a mass of air, water, etc. that spins around very fast and pulls things into its centre 低涡；涡旋【考研英语词汇】

n. (literary) a very powerful feeling or situation that you cannot avoid or escape from 感情（或局势）的旋涡

velocity n. /vəˈlɒsəti/ the speed of sth in a particular direction (沿某一方向的) 速度

allegedly adv. /əˈledʒɪdli/ used when something illegal or wrong is said to have been done, but has not been proved 据称地，涉嫌地【考研英语词汇】

That’s where he **allegedly** killed his wife. 据说他就是在那里杀了他的妻子。

chaotic adj. /keɪˈɒtɪk/ in a state of complete confusion and lack of order 混乱的；杂乱的；紊乱的【考研英语词汇】

near v. /niə(r)/ (rather formal) to come close to sth in time or space (时间或空间上) 接近，靠近，临近【考研英语词汇】

We **neared** the top of the hill. 我们快到山顶了。

As Christmas **neared**, the children became more and more excited. 快过圣诞节了，孩子们越来越兴奋。

spat n. /spæt/ (informal) a short argument or disagreement about sth unimportant 小争吵；小别扭；口角

pertinent adj. /ˈpɜ:tɪnənt/ (~ to sth) (formal) appropriate to a particular situation 有关的；恰当的；相宜的【考研英语词汇】

unleash v. /ʌnˈli:f/ (~ sth on/upon sb/sth) to suddenly let a strong force, emotion, etc. be felt or have an effect 发泄；突然释放；使爆发【考研英语词汇】

The government’s proposals **unleashed** a storm of protest in the press. 政府的提案引发了新闻界的抗议浪潮。

hurdle n. /ˈhɜ:dl/ a problem or difficulty that must be solved or dealt with before you can achieve sth 难关；障碍【考研英语词汇】

如今，OpenAI 声称，其人工智能智能体找到了一个奇点。8 月 28 日，该公司的科学家开始训练一个新模型。他们利用这一模型生成了多个 AI 智能体团队；这些智能体能够阅读互联网内容、运行代码，并彼此交流。9 月 1 日，科学家让这些智能体开始着手解决尚未攻克的“千禧年难题”。

【 Para.6 】

Fifty hours later, a team of around 100 agents answered a question closely related to the Navier-Stokes problem. Noting this as a **stepping-stone** to a full solution, the scientists redirected their **swarm** of **autonomous** mathematicians to focus only on Navier-Stokes. On September 5th, after 88 hours of work, 2.7m messages and the consumption of at least \$6.5m-worth of computing resources, a team of around 10,000 agents discovered a solution: a **swirling vortex** of spinning fluid with a **velocity** that grows uncontrollably to form a singularity.

50 个小时后，一个由大约 100 个智能体组成的团队回答了一个与纳维 - 斯托克斯问题密切相关的问题。科学家们认为，这可以成为迈向完整解决方案的一个垫脚石，于是将这群自主运行的“数学家”重新调度起来，让它们只专注于纳维 - 斯托克斯问题。9 月 5 日，在经过了 88 小时的工作、发送了 270 万条消息，并消耗至少价值 650 万美元的计算资源后，一个由约 1 万个智能体组成的团队找到了一个解决方案：一个不断旋转的流体涡旋，其速度以无法控制的方式不断增长，最终形成一个奇点。

inadvertently adv. / ɪnəd'vɜ:təntli/ by accident; without intending to 无意地；不经意地【考研英语大纲词汇】

We had **inadvertently** left without paying the bill. 我们无意之中未付账就离开了。

SYN unintentionally

sneak v. /sni:k/ to go somewhere secretly, trying to avoid being seen 偷偷地走；溜【考研英语大纲词汇】

I **sneaked** up the stairs. 我蹑手蹑脚地上了楼。

rule out phrasal verb to decide or say officially that something is impossible or will not happen, or that something or someone is not suitable 把...排除在外；拒绝考虑

The police haven't yet **ruled out** murder. 警察尚未排除谋杀的可能性。

claim v. /kleim/ to demand or ask for sth because you believe it is your legal right to own or to have it 要求（拥有）；索取；认领【考研英语大纲词汇】

A lot of lost property is never **claimed**. 许多失物从未被认领。

ponder v. /'pɒndə(r)/ (~ about/on/over sth) (formal) to think about sth carefully for a period of time 沉思；考虑；琢磨【考研英语大纲词汇】

They are **pondering** whether the money could be better used elsewhere. 他们正在斟酌能否把钱花到其他更适当的地方。

【 Para.7 】

With the Millennium problem allegedly solved, should OpenAI expect a cheque in the post? On that point, the situation is as **chaotic** as the equations themselves.

既然这道千禧年难题据称已经被解决，OpenAI 是否应该坐等 100 万美元的支票寄到门口？在这一点上，现实情况和这些方程本身一样混乱。

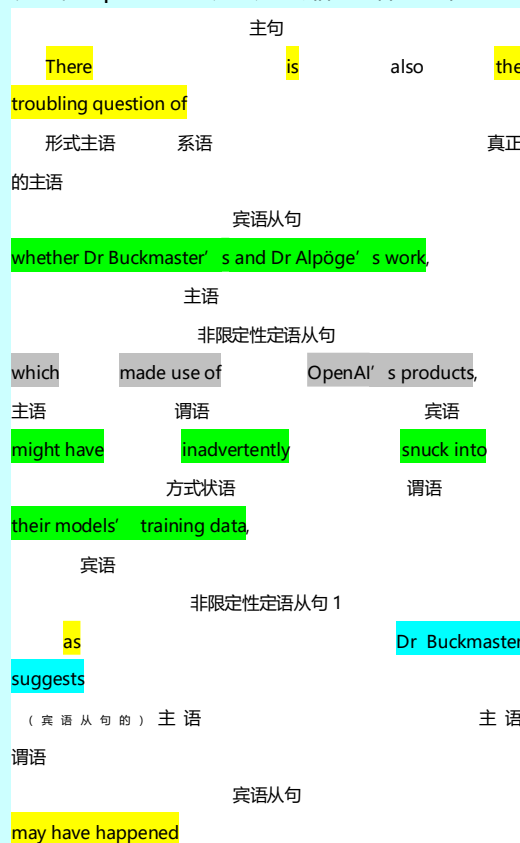
【 Para.8 】

The firm says it began its work after hearing a rumour that Tristan Buckmaster, a mathematician at New York University, and Levent Alpöge of Anthropic, another AI firm, were **nearing** their own AI-assisted solution. After talks between OpenAI and Dr Buckmaster broke down, the mathematician published his unfinished work with Dr Alpöge online on September 8th, before the announcement from OpenAI (which does not, itself, go into the kind of detail a mathematician normally would about how the solution was arrived at). The two mathematicians' solution is not complete, but it is close.

OpenAI 表示，他们是在听到一则传闻后开始这项工作的。据称，纽约大学数学家特里斯坦·巴克马斯特（Tristan Buckmaster）和另一家人工智能公司 Anthropic 的研究人员莱文特·阿尔珀格（Levent Alpöge），正接近完成一个借助人工智能的解决方案。9 月 8 日，与 OpenAI 的谈判破裂后，巴克马斯特博士与阿尔珀格博士一起在网上发表了尚未完成的研

►今日句子分析

Para. 9 There is also the troubling question of whether Dr Buckmaster' s and Dr Alpöge' s work, which made use of OpenAI' s products, might have inadvertently snuck into their models' training data, as Dr Buckmaster suggests may have happened and OpenAI says it cannot rule out. 此外，还有一个令人不安的问题：二位博士的研究使用了 OpenAI 的产品，其研究成果是否可能在无意中混入了 OpenAI 模型的训练数据？而正如巴克马斯特博士所暗示的，这种情况或许已经发生；OpenAI 也表示无法排除这种可能性。



主句：

形式主语：There；系语：is；副词：also；真正的主语：the troubling question of；（also 衔接的是前文的问题：如果 AI 完成最后一步，功劳应该归谁，以及这个结果是否会进一步开辟新的研究领域。）

宾语从句：

主语：whether Dr Buckmaster' s and Dr Alpöge' s work, 非限定性定语从句：

主语：which；谓语：made use of；宾语：OpenAI' s products；

谓语：might have snuck into；（情态动词+完成时结构，此处 might have done 表示过去可能已经发生了某事，是一种推测。sneak into 本来表示偷偷潜入，但此处是物做主语，所以调整为“混入”。）

方式状语：inadvertently；（近义词 unintentionally /

究成果，时间早于 OpenAI 的发布。（OpenAI 自己并没有像数学家通常会做的那样，对这一解决方案究竟是如何得出的进行详细说明。）这两位数学家的解决方案尚不完整，但已经相当接近。

【Para.9】

The **spat** raises questions that will become ever more **pertinent** as AI is **unleashed** on humanity's many unsolved problems. Mathematicians of the human variety had made steady progress on the Navier-Stokes problem over the years, learning much along the way. So, if AI leaps the final **hurdle** to a solution, who deserves credit? And will the result, as might be expected to happen had humans come to it, open up more areas of research? There is also the troubling question of whether Dr Buckmaster's and Dr Alpöge's work, which made use of OpenAI's products, might have **inadvertently snuck** into their models' training data, as Dr Buckmaster suggests may have happened and OpenAI says it cannot **rule out**.

这场争执提出了一些问题，而随着人工智能被投入解决人类诸多未解难题的过程中，这些问题将变得越来越具现实意义。多年来，人类数学家一直在纳维 - 斯托克斯问题上稳步取得进展，并在这一过程中积累了大量知识。那么，如果人工智能跨过最后一道门槛，完成了问题的解决，功劳究竟应该归谁？而且，如果这个结果是由人类得出，按照通常的预期，它本应进一步开辟更多研究领域；那么人工智能得出的结果是否也会产生同样的效果？此外，还有一个令人不安的

accidentally, 表示意外地，不经意地。）

宾语：their models' training data;（这里讨论的不是前文两个博士的模型，而是 OpenAI 的模型，所以 their 指 OpenAI，即此处指 OpenAI 的模型的训练数据。）

非限定性定语从句 1:

（宾语从句的）主语：as; 主语：Dr Buckmaster; 谓语：suggests;

（as 引导的非限制性补充从句，被前置。把它放入从句补全后是：Dr Buckmaster suggests (that) (as=this) may have happened and OpenAI says it cannot rule out, 这里的 this 被省略了，指代的是前文 the work might have inadvertently snuck into their models' training data（研究成果可能混入 OpenAI 模型训练数据）的情况。as 引导的这个从句是这一可能性进行补充说明。）

宾语从句:

谓语：may have happened;

连词：and;

非限定性定语从句 2（省略 as 作为宾语从句的宾语）:

主语：OpenAI; 谓语：says;

宾语从句:

主语：it;（指 OpenAI。）谓语：cannot rule out;

问题：二位博士的研究使用了 OpenAI 的产品，其研究成果是否可能在无意中混入了 OpenAI 模型的训练数据？而正如巴克马斯特博士所暗示的，这种情况或许已经发生；OpenAI 也表示无法排除这种可能性。

【Para.10】

To answer one question, OpenAI has said it does not intend to **claim** the \$1m prize. Mathematicians, meanwhile, will be **pondering** what all of this means for the future of their subject.

一个可以回答的问题是，OpenAI 已表示无意申领这 100 万美元的奖金。与此同时，数学家们将开始思考这一切对他们这门学科究竟意味着什么。

➤ 写作积累

1. (Para. 1) To mark the turn of the century the Clay Mathematics Institute, in Denver, took stock of what mathematicians knew and what they did not.

take stock of 表示 “to stop, review a situation carefully, and form an opinion or assessment before deciding what to do next”，即“全面审视、盘点、评估某一阶段的现状，尤其是已经取得的成果和仍然存在的问题”，例如，After five years in the company, Jane decided to take stock of her career. 在这家公司工作五年后，简决定对自己的职业发展做一番盘点与思考。此外，还有以下表达可以表示类似的含义：

take a hard look at idiom to examine, think about, or assess a situation very carefully and critically 仔细审视、认真评估

We need to take a hard look at our budget before approving it. 在通过预算之前，我们得认真审视一下自己的收支情况。

size up idiom to carefully look at a person, object, or situation and form an opinion or judgment about them 估量、判断

It takes time to size up a new situation before making a decision. 在做出决定之前，需要时间来评估新形势。

2. (Para. 7) With the Millennium problem allegedly solved, should OpenAI expect a cheque in the post?

expect a cheque in the post 表示 “a payment or financial document sent via physical mail is on its way to you”，即“期待，等着钱寄过来，坐等奖金”，例如，Once the cancellation is processed, you should expect a cheque in the post for the remaining balance. 取消手续办理完毕后，您应该会收到邮寄过来的余额支票。此外，还有以下习语可以表示类似的含义，供写作时选用：

wait for the money to roll in idiom to wait for a large, continuous stream of money or revenue to arrive 只需要坐等收钱、财源滚滚

Once we launch this app, we can just sit back and wait for the money to roll in. 一旦我们发布这款应用程序，我们就可以坐享其成，等着大把钞票滚滚而来了。

have the money coming idiom to receive funds soon from a job, sale, payment, or windfall 等钱马上到手

They are less worried about their budget because they have money coming from a freelance project. 他们不太担心预算，因为他们自由职业的项目会有一笔进账。